

Q. The Bharat Ball Bearings Ltd., has collected the following data.

12, 19, 21, 30, 13, 19, 22, 31, 17, 20, 24, 31, 18, 21, 27, 31.

Compute the arithmetic mean, the median and the mode using the sixteen observations given.

Ans: A.M. = 22.25 , Median = 21, Mode = 31

Q. Calculate mean, median and mode from the following data of the heights in inches of a group of students:

61, 62, 63, 61, 63, 64, 64, 60, 65, 63, 64, 65, 66, 64

Ans: $M = 63.2$, $Md = 63.5$, $Mo = 64$

Fill in the blanks:

1. can be calculated from a frequency distribution with open end classes.
2. In the calculation of, all the observations are taken into consideration.
3. Average rainfall of a city from Monday to Saturday is 0.3 inch. Due to heavy rainfall on Sunday, the average rainfall for the week increases to 0.5 inch. The rainfall on Sunday was ...
4. The sum of squared deviations is minimum when taken from
5. The sum of absolute deviations is minimum when taken from
6. Median = Quartile.
7. Median is the average suited for classes.
8. For studying phenomenon like intelligence and honesty is a better average to be used while for phenomenon like size of shoes or readymade garments the average to be preferred is

9. Typist A can type a sheet in 5 minutes, typist B in 6 minutes and typist C in 8 minutes. The average number of sheets typed per hour per typist is

10. The mean of 10 observations is 20 and median is 15. If 5 is added to each observation, the new mean is and median is

11. A distribution with two modes is called and with more than two modes is called

12. Average suited for a qualitative phenomenon is

13. Relationship between Md, Q1, Q2 and Q3 is

14. The empirical relationship between mean, median and mode for a moderately asymmetrical distribution is

15. If A.M. of two numbers is 17 and G.M. is 15, find the H.M. of these numbers.

Answers: (Md = Median, Mo = Mode)

1. Median or Mode
2. Mean
3. 1.7 inches
4. Mean
5. Median
6. Second
7. Open end
8. Median, Mode
9. 9.83 sheets/hr
10. 25, 20
11. Bi-modal, Multi-modal
12. Median
13. $Q1 \leq Q2 = Md \leq Q3$
14. $Mode = 3Md - 2M.$
15. 13.24

MEASURES OF ASYMMETRY

Q. The sum of 50 observations is 500 and the sum of their squares is 6,000 and median is 12. Compute the coefficient of variation and the coefficient of skewness.

Ans:

1. Coefficient of Variation (C.V.) = 44.7
2. Karl Pearson's coefficient of skewness = -1.34 .

Q. The following information was obtained from the records of a factory relating to the wages. Arithmetic Mean: Rs. 56.80;

Median: Rs. 59.50; S.D.: Rs. 12.40

Compute the coefficient of variation, the coefficient of skewness and Mode

Ans:

1. Coefficient of Variation (C.V.) = 21.83
2. Karl Pearson's coefficient of skewness = -0.65
3. Hence, the distribution of wages is _____ (positively/negatively) skewed i.e., it has a _____ (shorter/ longer) tail towards the _____ (right/ left)
4. $M_o = 3M_d - 2M = \text{Rs.}64.90$

Fill in the blanks:

- (i) If $\beta_2 = 3$, the curve is called
- (ii) If $\beta_2 > 3$, the curve is called
- (iii) If $\beta_2 < 3$, the curve is called
- (iv) If $\beta_1 = 0$, the curve is called

Ans:

- (i) Mesokurtic, (ii) Leptokurtic, (iii) Platykurtic, (iv) Symmetrical

Fill in the blanks:

- (i) Literal meaning of skewness is “.....”
- (ii) Kurtosis is a measure of of the frequency curve.
- (iii) For a symmetrical distribution, mean, median and mode
- (iv) If Mean < Mode, the distribution is skewed.
- (v) If Mean > Median, the distribution is skewed.
- (vi) Bowley's coefficient of skewness lies between and
- (vii) $\beta_1 = 0$ implies that distribution is
- (viii) If $\gamma_1 > 0$, the distribution is skewed
- (ix) If $\gamma_1 < 0$, the distribution is skewed.
- (x) For a normal curve β_2
- (xi) If $\beta_2 > 3$, the curve is called
- (xii) If $\beta_2 < 3$, the curve is called
- (xiii) An absolute measure of skewness based on moments is

- (xiv) An absolute measure of skewness based on quartiles is
- (xv) Relative measure of skewness based on mean, SD. and mode is
- (xvi) Relative measure of kurtosis is
- (xvii) Relative measure of skewness in terms of moments is
- (xviii) For a moderately asymmetrical distribution : Mean – Median = ? (Mean – Mode)
- (xix) If $\mu_3 = -1.48$, the curve of the given distribution is stretched more to thethan to the
- (xx) For a symmetrical distribution, are equidistant from
- (xxi) Mean = First moment about
- (xxii) Variance = moment about mean.
- (xxiii) In a moderately asymmetrical distribution the distance between ... and ... is ... the distance between ... and

Ans:

- (i) Lack of symmetry,
- (ii) Convexity (flatness or peakedness),
- (iii) Coincide,
- (iv) Negatively,
- (v) Positively,
- (vi) -1 and 1,
- (vii) Symmetrical,
- (viii) Positively,

- (ix) Negatively,
- (x) 3,
- (xi) Lepto-kurtic,
- (xii) Platy-kurtic,
- (xii) μ_3 ,
- (xiv) $(Q_3 - M_d) - (M_d - Q_1)$,
- (xv) $(M - M_o)/\sigma$,
- (xvi) β_2 or γ_2 ,
- (xvii) β_1 or γ_1 ,
- (xviii) $1/3$,
- (xix) Left, Right,
- (xx) Quartiles, median,
- (xxi) Origin,
- (xxii) Second
- (xxiii) Mean, Mode, Thrice, Mean, Median.

MEASURES OF DISPERSION

Calculate the range and the coefficient of range of A's monthly earnings for a year.

Month Earnings (Rs.)

1	13900
2	15000
3	15100
4	15100
5	15700
6	15800
7	16000
8	16100
9	16200
10	16200
11	17300
12	17500

Ans: Range = Rs. 3,600 , Coefficient of range = 0.115

The following table gives the age distribution of a group of 50 individuals.

Age (in years) No. of persons

16 – 20	10
21 – 25	15
26 – 30	17
31 – 36	8

Range = 20 years, Coefficient of range = 0.39.

Find (i) Inter-quartile range, (ii) Semi-inter-quartile range, and (iii) Coefficient of quartile deviation, from the following frequency distribution:

Marks :	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of students :	60	45	120	25	90	80	120	60

$$\text{semi-interquartile range} = \frac{Q_3 - Q_1}{2}$$

Ans: (i) 38.75, (ii) 19.375, (iii) 0.3647.

Fill in the blanks:

- Algebraic sum of deviations is zero from
- The sum of absolute deviations is minimum from
- Standard deviation is always than range.
- Standard deviation is always than mean deviation.
- The mean and SD of 100 observations are 50 and 10 respectively.
The new:
 - Mean =, SD =, if 2 is added to each observation.
 - Mean =, SD =, if 3 is subtracted from each observation.
 - Mean = ..., SD = ..., if each observation is multiplied by 5
 - If 2 is subtracted from each observation and then it is divided by 5.
- Variance is the value of mean square deviation.
- If $Q_1 = 10$, $Q_3 = 40$, the coefficient of quartile deviation is
- If 25% of the items in a distribution are less than 10 and 25% are more than 40, the coefficient of quartile deviation (CQD) is
- The median and SD of a distribution are 15 and 5 respectively. If each item is increased by 5, the new median = and SD =
- A computer showed that the SD of 40 observations ranging from 120 to 150 is 35.

11. Quartile deviation is measure of dispersion.

Ans:

1. Arithmetic mean
2. Median
3. Less or equal
4. Greater
5. (a) 52, 10 (b) 47, 10 (c) 250, 50 (d) 9.6, 2
6. Minimum
7. $CQD = 0.6$
8. 15
9. 20, 5
10. Wrong, since SD. can't exceed range.
11. Absolute

For Question No. 5 & 9 use below

1. If a constant k is added/subtracted to every observation:

- New mean = old mean $\pm$ k
- Standard deviation = unchanged

2. If every observation is multiplied/divided by k:

- New mean = old mean $\times$ k
- New standard deviation = old standard deviation $\times |k|$

3. Adding or subtracting a constant:

- New median = old median $\pm$ constant
- SD = unchanged

Match the correct parts to make a valid statement:

- | | |
|--|---|
| (a) Algebraic sum of deviations from mean | (i) $Q3 - Q1$ |
| (b) Coefficient of Mean Deviation | (ii) $100 \sigma \text{ Mean}$ |
| (c) Variance | (iii) Zero |
| (d) Quartile Deviation | (iv) $\frac{1}{N} \sum f X - \bar{X} $ |
| (e) Coefficient of Variation | (v) $\frac{1}{N} \sum f (X - \bar{X})^2$ |
| (f) Sum of absolute deviations from median | (vi) $\frac{\text{M.D. about Mean}}{\text{Mean}}$ |
| (g) Interquartile range | (vii) $(Q3 - Q1)/2$ |
| (h) Mean deviation | (viii) Minimum |

Ans:

- (a) — (iii); (b) — (vi); (c) — (v); (d) — (vii); (e) — (ii); (f) — (viii); (g) — (i);
(h) — (iv)
-

Select the correct answer option:

(i) Algebraic sum of deviations from mean is :

(a) Positive, (b) Negative, (c) Zero, (d) Different for each case

(ii) Sum of squares of deviations is minimum when taken from

(a) Mean, (b) Median, (c) Mode, (d) None of these.

(iii) Sum of absolute deviations is minimum when measured from:

(a) Mean, (b) Median, (c) Mode, (d) None of these.

(iv) For a discrete frequency distribution:

(a) $S.D. \leq M.D.$, (b) $S.D. \geq M.D.$, (c) $S.D. > M.D.$, (d) $S.D. < M.D.$ (e) None of these, where M.D. = Mean Deviation from mean.

(v) The range of a given distribution is

(a) greater than SD (b) Less than SD (c) Equal to SD (d) None of these.

(vi) The measure of dispersion independent of frequencies of the given distribution is

(a) Range, (b) SD (c) M.D., (d) Q.D.

(vii) In case of open end classes, an appropriate measure of dispersion to be used is

(a) Range, (b) Q.D., (c) M.D., (d) SD

(viii) Measure of dispersion which is affected most by extreme observations is.

(a) Range, (b) Q.D., (c) M.D., (d) SD

(ix) Mean deviation from median (MD) is given by

(a) $\frac{\sum |X - Md|}{n}$, (b) $\frac{\sum |X - Md|}{\sqrt{n}}$, (c) $\frac{\sqrt{\sum |X - Md|^2}}{n}$ (d) $\frac{\sum |X - Md|^2}{n}$

(x) Quartile deviation is given by:

(a) $Q3 - Q1$, (b) $Q3 - Q2 / 2$, (c) $Q2 - Q1 / 2$, (d) $Q3 - Q1 / 2$.

(xi) If the distribution is approximately normal, then

(a) $M.D. = \frac{2}{5} \sigma$, (b) $M.D. = \frac{3}{5} \sigma$, (c) $M.D. = \frac{4}{5} \sigma$, (d) None of these.

(xii) For a normal distribution,

(a) $Q.D. = \frac{1}{3} \sigma$, (b) $Q.D. = \frac{2}{3} \sigma$, (c) $Q.D. = \sigma$, (d) None of these.

(xiii) For a normal distribution,

(a) $Q.D. > M.D.$, (b) $Q.D. < M.D.$, (c) $Q.D. = M.D.$ (d) None of these.

(xiv) For a normal distribution, the range $\text{mean} \pm 1.\sigma$ covers

(a) 65%, (b) 68.26%, (c) 85%, (d) 95% of items

Ans:

- (i)— (c);
 - (ii) — (a);
 - (iii) — (b);
 - (iv)— (b);
 - (v)— (a);
 - (vi)— (a);
 - (vii) — (b);
 - (viii) — (a);
 - (ix)— (a);
 - (x)— (d);
 - (xi) — (c);
 - (xii) — (b);
 - (xiii)— (b);
 - (xiv)— (b).
-

(a) If SD of a group is 15, find the most likely value of

(i) Mean deviation and (ii) Quartile deviation of that group.

(b) If mean deviation of a distribution is 20, find the most likely value of

(i) SD and (ii) Q.D.

(c) If quartile deviation of a distribution is 6, find the most likely value of
(i) SD and (ii) M.D.

(d) If quartile deviation of a distribution is 20 and its mean is 60, obtain the most likely value of

(i) Coefficient of variation, (ii) Mean deviation and (iii) Coefficient of mean deviation.

NOTE : In all the above parts, assume that the distribution is normal.

Ans:

(a) M.D. = 12, Q.D. = 10, (b) SD = 25, Q.D. = 16.67, (c) SD = 9, M.D. = 7.2,
(d) C.V. = 50, M.D. = 24, Coefficient of M.D. = 0.4

For Question (d) –

$$SD = 3/2 \times QD$$

$$= 3/2 \times 20$$

$$= 30$$

$$SD = (5/4) \times MD$$

$$30 = 5/4 \times MD$$

$$MD = 24$$

$$\text{Coefficient of MD} = \frac{MD}{\text{Mean}}$$

$$= 24/60$$

$$= 0.4$$

$$\text{Coefficient of variation} = \frac{SD}{\text{Mean}}$$

$$= 30/60$$

$$= 1/2$$

$$= 0.5 \text{ or } 50\%$$

SUNBEAM

Q. A uniform dice is thrown at random. Find the probability that the number on it is:

(i) 5, (ii) greater than 4, (iii) even.

Ans:

(i) $\frac{1}{6}$ (ii) $\frac{1}{3}$ (iii) $\frac{1}{2}$

Q. In a single throw with two uniform dice find the probability of throwing (i) Five, (ii) Eight.

Ans:

(i) $\frac{1}{9}$ (ii) $\frac{5}{36}$

Q. Four cards are drawn at random from a pack of 52 cards. Find the probability that

(i) They are a king, a queen, a jack and an ace.

(ii) Two are kings and two are aces.

(iii) All are diamonds.

(iv) Two are red and two are black.

(v) There is one card of each suit.

(vi) There are two cards of clubs and two cards of diamonds.

Ans:

(i) $\frac{256}{52_4C}$

(ii) $\frac{36}{52_4C}$

(iii) $\frac{715}{52_4C}$

$$\begin{aligned} \text{(iv)} \quad & \frac{{}^{26}_2C \times {}^{26}_2C}{{}^{52}_4C} \\ \text{(v)} \quad & \frac{{}^{13}_1C \times {}^{13}_1C \times {}^{13}_1C \times {}^{13}_1C}{{}^{52}_4C} \\ \text{(vi)} \quad & \frac{{}^{13}_2C \times {}^{13}_2C}{{}^{52}_4C} \end{aligned}$$

Q. A bag contains 20 tickets marked with numbers 1 to 20. One ticket is drawn at random. Find the probability that it will be a multiple of (i) 2 or 5, (ii) 3 or 5.

Ans:

- (i) $3/5$ Or 0.6
- (ii) $9/20$ or 0.45

Q. An bag contains 8 white and 3 red balls. If two balls are drawn at random, find the probability that:

- (i) both are white, (ii) both are red, (iii) one is of each color.

Ans:

- (i) $28/55$
- (ii) $3/55$
- (iii) $24/55$

Q. A bag contains 7 white and 9 black balls. Two balls are drawn in succession at random. What is the probability that one of them is white and the other is black?

Ans. $21/40$.

Q. A bag contains eight balls, five being red and three white. If a man selects two balls at random from the bag, what is the probability that he will get one ball of each colour?

Ans. $15/28$

Q. A bag contains 5 white and 3 black balls. Two balls are drawn at random one after the other without replacement. Find the probability that balls drawn are black.

Ans. $3/28$

Q. A bag contains 4 white, 5 red and 6 green balls. Three balls are drawn at random. What is the probability that a white, a red and a green ball are drawn?

Ans. $24/91$.

Q. A bag contains 8 black, 3 red and 9 white balls. If 3 balls are drawn at random, find the probability that (a) all are black, (b) 2 are black and 1 is white, (c) 1 is of each color, (d) the balls are drawn in the order black, red and white, (e) None is red.

Ans. (a) $14/285$, (b) $21/95$, (c) $18/95$, (d) $3/95$, (e) $34/57$.

Q. The Federal Match Company has forty female employees and sixty male employees. If two employees are selected at random, what is the probability that (i) both will be males, (ii) both will be females, (iii) there will be one of each sex?

Ans. (i) 0.357 , (ii) 0.157 , (iii) 0.4848 .

Q. Among the 90 pieces of mail delivered to an office, 50 are addressed to the accounting department and 40 are addressed to the marketing department. If two of these pieces of mail are delivered to the manager's office by mistake, and the selection is random, what are the probabilities that :

- (i) Both of them should have been delivered to the accounting department;
- (ii) Both of them should have been delivered to the marketing department;
- (iii) One should have been delivered to the accounting department and the other to be marketing department?

Ans.

(i) $\frac{{}^{50}C_2}{{}^{90}C_2} = 0.3059 ;$

(ii) $\frac{{}^{40}C_2}{{}^{90}C_2} = 0.1948 ;$

(iii) $\frac{{}^{50}C_1 \times {}^{40}C_1}{{}^{90}C_2} = 0.4994$

Q. If a single draw is made from a pack of 52 cards, what is the probability of securing either an ace of spades or a jack of clubs.

Ans. 1 / 26.

Q. A committee of four has to be formed from among 3 economists, 4 engineers, 2 statisticians and 1 doctor.

(i) What is the probability that each of the four professions is represented on the committee ?

(ii) What is the probability that the committee consists of the doctor and at least one economist ?

Ans:

(i) $4 / 35 = 0.1143$

(ii) $32 / 105 = 0.3048$